

Imperial College London

BSc/MSci EXAMINATION May 2016

This paper is also taken for the relevant Examination for the Associateship

PRINCIPLES OF INSTRUMENTATION

For 3rd/4th-Year Physics Students

Thursday, 19th May 2016: 10:00 to 12:00

The paper consists of two sections: A & B.

Section A contains one question, comprising small parts. [40 marks total]

Section B contains four questions on selected parts of the course. [30 marks each]

Candidates are required to:

*Answer **ALL** parts of Section A and **TWO** questions from Section B.*

Marks shown on this paper are indicative of those the Examiners anticipate assigning.

General Instructions

Complete the front cover of each of the THREE answer books provided.

If an electronic calculator is used, write its serial number at the top of the front cover of each answer book.

USE ONE ANSWER BOOK FOR EACH QUESTION.

Enter the number of each question attempted in the box on the front cover of its corresponding answer book.

Hand in THREE answer books even if they have not all been used.

You are reminded that Examiners attach great importance to legibility, accuracy and clarity of expression.

SECTION A

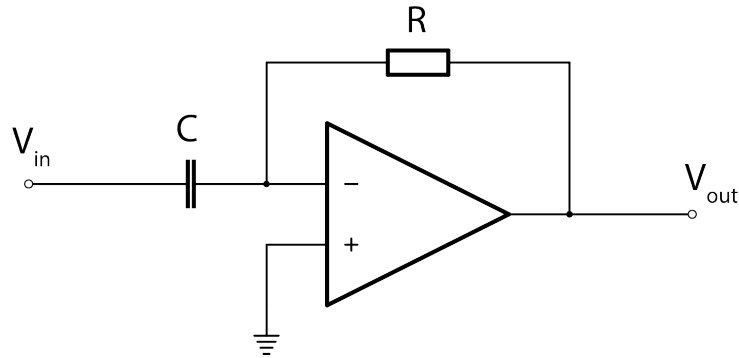
1. (i) (a) Briefly summarise the properties of the ideal operational amplifier. What are the two 'golden-rules' which simplify the analysis of such circuits?
- (b) The diagram below shows an op-amp which is assumed to be ideal. Show that

$$V_{out} = -RC \frac{d}{dt} V_{in}$$

- (c) With $R = 10 \text{ k}\Omega$ and $C = 100 \mu\text{F}$, sketch V_{out} for

$$V_{in}(t) = \begin{cases} a & t < -a \\ -t & -a \leq t \leq a \\ -a & t > a \end{cases}$$

[10 marks]



- (ii) A photomultiplier tube has 6 dynodes; each on average liberates an additional 9 electrons for every incident electron. The photo-cathode has a quantum efficiency of 25%. The anode is connected to ground by a 50Ω resistor across which the output voltage is measured.

- (a) If the cathode is illuminated with green light (wavelength 548 nm) at optical power 108 pW, find the output voltage, assuming no dark-current.
- (b) Assuming no flicker noise, what other processes will generate a noise voltage across the 50Ω resistor?

[10 marks]

- (iii) (a) Describe the frequency characteristics of flicker noise and explain why noise of this type is known as 'pink noise'.
- (b) The current in a thermionic diode is recorded over a period of six minutes and the flicker noise power is measured with a bandwidth of 1 kHz. After what period of time would the noise power be expected to double?

[8 marks]

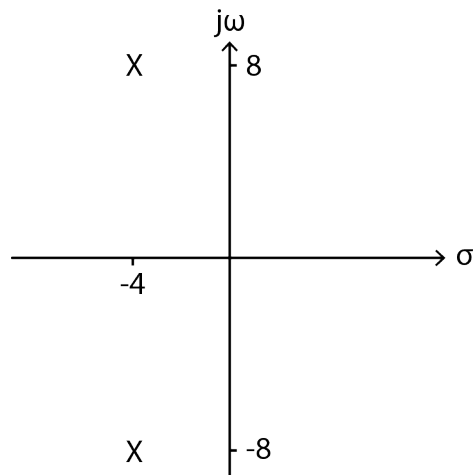
(iv) The diagram below shows the pole-zero plot for a linear time-invariant system with a static sensitivity $K_s = 8$.

(a) Show that the system may be represented by a differential equation of the form

$$640y = \ddot{x} + 8\dot{x} + 80x$$

where $y(t)$ is the input to the system and $x(t)$ the output.

(b) Find an expression for the system's impulse response.



[12 marks]

[Total 40 marks]

SECTION B

2. (i) A co-axial cable has a capacitance per metre $C' = 95 \text{ pF m}^{-1}$ and inductance per metre $L' = 0.2375 \mu\text{H m}^{-1}$.
- What is the meaning of the term 'characteristic impedance'? What is the numerical value for this cable?
 - A 100 m length of the cable has the central conductor joined to the grounded shield at the far end. A very short pulse of amplitude 5 V is introduced at the source end at time $t = 0$. At what time does the pulse reach the far end?
 - By considering the boundary conditions at the far end, deduce the amplitude and polarity of the reflected pulse.

[10 marks]

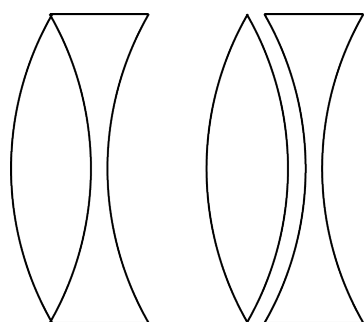
- (ii) A $2 V_{\text{RMS}}$ AC signal generator is connected to a variable load resistance R (ohms). The RMS power in watts developed in the load is determined to be

$$P = \frac{4R}{2.5 \times 10^3 + 10^2 R + R^2}$$

- Determine the signal generator's source resistance.
- Draw the circuit for a signal transformer which will couple maximum power from the signal generator into a load of 8Ω . Indicate the transformer turns ratio on the diagram.

[10 marks]

- (iii) Chromatic aberration caused by dispersion can be corrected by the use of a 'doublet' lens comprising two glasses of different refractive indices. The diagram below shows two types of doublet lenses. The cemented doublet (a) consists of a crown-glass ($n = 1.5$) convex element directly bonded to a flint-glass ($n = 1.8$) concave element. The oil-spaced doublet (b) comprises identical elements with a transparent oil filling the gap.



(a) Cemented

(b) Oil-spaced

- What should be the refractive index of the oil? Briefly explain the physical origin of your answer.

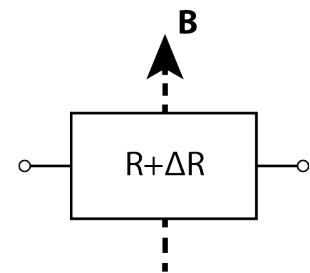
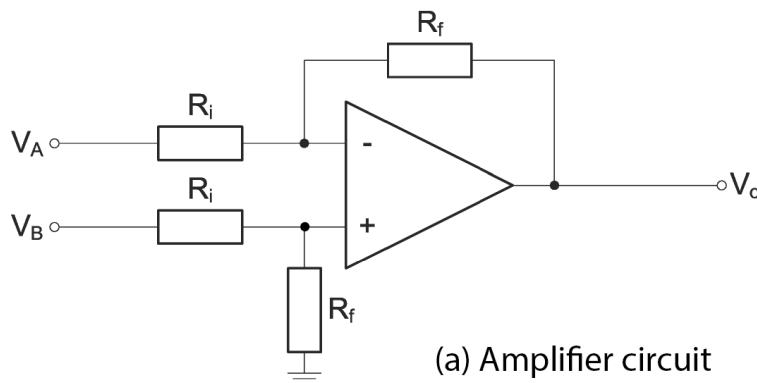
[This question continues on the next page ...]

- (b) Calculate the percentage improvement in optical power transmission achieved through using the oil-spaced doublet when compared to the cemented doublet.

[10 marks]

[Total 30 marks]

3. (i) Diagram (a) below shows an amplifier circuit which uses an operational amplifier which is assumed to have ideal properties. Find an expression for the output of the circuit. [8 marks]



- (ii) Diagram (b) shows a 'magnetoresistor' with resistance $R + \Delta R$ where $R = 1 \text{ k}\Omega$ is the zero-field resistance and ΔR is a positive or negative change in resistance in response to an applied positive or negative magnetic field B applied in the direction of the arrow. The static sensitivity of the device is defined by $\Delta R = 2 \times 10^5 \Omega/\text{T}$ subject to a maximum change in resistance of $\pm 20 \Omega$.
- Four magnetoresistors may be arranged as a Wheatstone bridge. Assume a bridge voltage of 5 V. Sketch the circuit and find an expression for the bridge output voltage as a function of applied magnetic field.
 - Suggest values for resistors R_f and R_i which will allow circuit (a) to amplify this voltage resulting in an output of 0.65 V at Earth's field (assumed to be $65 \mu\text{T}$).
 - For the amplified output, sketch the static sensitivity plot as a function of applied magnetic field over the range $\pm 200 \mu\text{T}$. Annotate your sketch with any important features.
 - The op-amp has, in open loop, a gain $A = 10^5$ and bandwidth of 10 Hz. What will be the bandwidth for magnetic field measurement?

[22 marks]

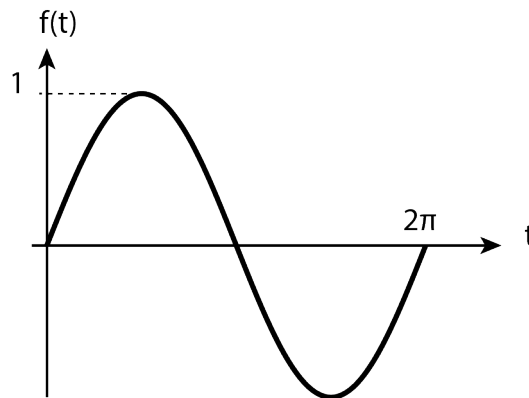
[Total 30 marks]

4. (i) (a) Give the integral form of the Laplace transform and use this to prove the transform's time-shift property:

$$\mathcal{L}\{f(t - \lambda)u(t - \lambda)\} = e^{-s\lambda}F(s)$$

- (b) The diagram below shows a single period of a sine-function starting at time $t = 0$. Write an expression for this waveform as a linear superposition of functions and hence deduce the waveform's Laplace transform.

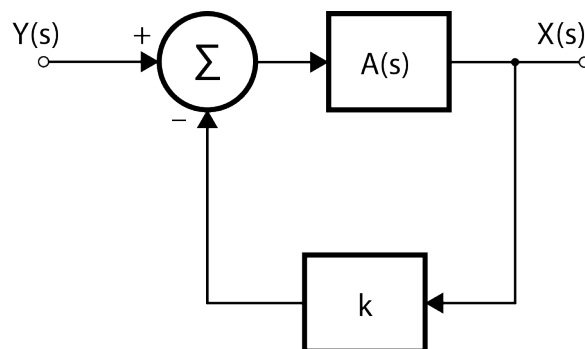
[10 marks]



- (ii) In the system diagram below the block labelled K represents an ideal amplifier of gain k where k may be any real constant.

- (a) Show that the transfer function of the system is

$$G(s) = \frac{A(s)}{1 + kA(s)}$$



- (b) A has a transfer function

$$A(s) = \frac{1}{s^2 + s - 2}$$

Is A stable with respect to an arbitrary perturbation at its input? Explain your reasoning. Find an expression for A 's impulse response and indicate whether this supports your answer.

- (c) Find the range of values of k for which the system is either stable or unstable.
- (d) Classify, as a function of k , the possible types of impulse response which the system may exhibit in the stable range.

[20 marks]

[Total 30 marks]

5. (i) A source provides a sinusoidal voltage amplitude 240 V and frequency 50 Hz.
- (a) Give the diagram of a circuit, including an ideal diode-bridge, to convert this to a full-wave rectified signal of amplitude 5 V.
- (b) Sketch the bridge output and show that it is represented by

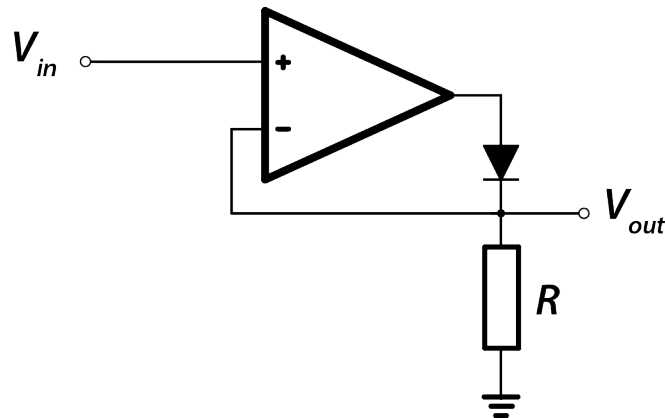
$$f(t) = \frac{10}{\pi} + \sum_{n=2,4,6\dots}^{\infty} \frac{20 \cos 100n\pi t}{\pi(1 - n^2)}$$

- (c) Discuss whether the circuit is linear time-invariant (LTI).
- (d) Sketch the bridge output signal again, now assuming the diodes to be real-world and to have a forward voltage drop of $V_d = 0.6$ V.

[20 marks]

- (ii) The circuit below is known as the precision rectifier or 'super-diode'. With reference to the properties of real-world op-amps and diodes, explain the function of this circuit. For an op-amp open-loop gain A and diode with forward voltage drop V_d , write an expression for the circuit's apparent forward voltage drop.

[10 marks]



[Total 30 marks]

You may use

$$f(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} (A_n \cos n\omega t + B_n \sin n\omega t)$$

$$A_n = \frac{2}{T} \int_{t_0}^{t_0+T} f(t) \cos n\omega t dt$$

$$B_n = \frac{2}{T} \int_{t_0}^{t_0+T} f(t) \sin n\omega t dt$$

Note that for all odd functions the coefficients B_n exist while the A_n are zero, and vice-versa for even functions.

Table of Laplace Transforms

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
t^n for $n > 0$	$\frac{n!}{s^{(n+1)}}$
e^{-at}	$\frac{1}{s+a}$
te^{-at}	$\frac{1}{(s+a)^2}$
$\frac{1}{b-a}(e^{-at} - e^{-bt})$	$\frac{1}{(s+a)(s+b)}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$e^{-at} \sin(\omega t)$	$\frac{\omega}{(s+a)^2 + \omega^2}$
$e^{-at} \cos(\omega t)$	$\frac{s+a}{(s+a)^2 + \omega^2}$